Deep-sub-cycle ultrafast optical pulses

Hongliang Dang,* Jiaxin Gao,* Hao Wu, Yudong Zhang, Xin Guo, and Limin Tong†
New Cornerstone Science Laboratory, State Key Laboratory of Extreme Photonics and Instrumentation, College of Optical Science and Engineering, Zhejiang University, Hangzhou 310027, China

(Received 19 January 2025; accepted 17 June 2025; published 10 July 2025; corrected 29 July 2025)

Subcycle optical pulse is of great importance for ultrafast science and technology. While a narrower pulse can offer a higher temporal resolution, so far the pulse width has not reached the limit of half an optical cycle. Here we propose to generate deep-sub-cycle (i.e., less than half a cycle, sub-half-cycle) ultrafast optical pulses that break the half-cycle limit via inverse Compton scattering of a deep-subwavelength-confined THz optical driving field by relativistic electrons. Quantitatively, based on a deep-subwavelength-confined 0.4-THz 1-ps pulsed driving field, we obtain a 0.095-cycle 3.6-fs-width mid-infrared femtosecond pulse with a peak frequency of ∼26 THz (∼38 fs in optical cycle) by using a 3-MeV 1-fs electron bunch, and a 0.091-cycle 180-as-width visible attosecond pulse with a peak frequency of ∼0.49 PHz (∼2.0 fs in optical cycle) by using a 15-MeV 50-as electron bunch, respectively. Below the optical damage threshold of the sapphire prisms used for field confinement, the single-pulse photon number can exceed 10^5. Such pulses may open opportunities for studying light-matter interaction on the deep-sub-cycle level, and pave a way to unprecedented optical technology ranging from temporal super-resolution optical microscopy and spectroscopy to unconventional atom/molecule polarization and manipulation.

DOI: 10.1103/s269-kl9z

I. INTRODUCTION

Space and time are basic scales for understanding the world. Higher spatial (e.g., from optical [1,2] to electron microscopy [3,4]) and temporal (e.g., from femtosecond [5,6] to attosecond [7,8] technology) resolutions are always desired for a better understanding of the microscopic world on a deeper level [5,8]. On the temporal scale, so far the highest resolution comes from ultrafast optical technology, which has been widely employed for studying the dynamics of molecular or atomic processes from femtosecond to attosecond scales [3,5–10]. Typically, to pursue a higher temporal resolution, we have to resort to shorter optical pulses with higher photon energy (i.e., higher frequency or shorter free-space wavelength), which is analogous to achieving higher diffraction-limited spatial resolution by reducing the free-space wavelength of the probing light (e.g., x-ray microscopy [11]). However, for a certain light-matter interaction, reducing the free-space wavelength (i.e., increasing the light frequency) far away from the intrinsic resonance of the matter will significantly reduce the excitation efficiency (i.e., the matter becomes "transparent" to the probing light). Therefore, a more effective way to increase the spatial resolution and resolve the subdiffraction structure is not solely by reducing the wavelength, but by confining the light field to a spatial scale well below the half wavelength (i.e., deep-subwavelength scale) that breaks the optical diffraction limit in the near field, and obtain a super-diffraction-limit optical resolution at an optical frequency not much far away from the concerned resonance of the material, such as optical near-field microscopy [1,12,13]. Compared with the spatial super-resolution technology that has been well developed, the temporal super-resolution remains challenging, due to the absence of the deep-sub-cycle optical field with a pulse width shorter than half a cycle.

In recent years, various approaches [14–21] based on, e.g., optical parametric chirped pulse amplification (OPCPA) [14,15], optical arbitrary wave-form generation (OAWG) [16,17], and relativistic electron sheet (RES) [18,19], have been developed to generate subcycle ultrafast optical pulses. However, due to difficulties such as limited gain bandwidth of femtosecond laser sources [20] and insufficient degree of freedom of wave-form control [21], the narrowest pulse width obtained remains larger than $T_c/2$ ($T_c/2$ on the temporal scale is analogous to the diffraction limit of $\lambda/2$ on the spatial scale, where T_c and λ are the cycle and the wavelength of the light).

II. BASIC PRINCIPLE FOR BREAKING THE HALF-CYCLE LIMIT VIA ICS

Here, we propose to generate deep-sub-cycle ultrafast optical pulses via inverse Compton scattering (ICS) in a deep-sub-wavelength-confined optical field, in which the field with a spatial full width at half maximum ($\mathrm{FWHM_S}$) smaller than half a wavelength is converted into a deep-sub-cycle optical pulse with a temporal full width at half maximum ($\mathrm{FWHM_T}$) smaller than half a cycle by relativistic electrons passing through.

Our approach to breaking the half-cycle limit of an optical pulse is illustrated in Fig. 1(a), when a photon (with a frequency ν_0 and wavelength λ_0) collides with an electron

*These authors contributed equally to this work.
†Contact author: phytong@zju.edu.cn

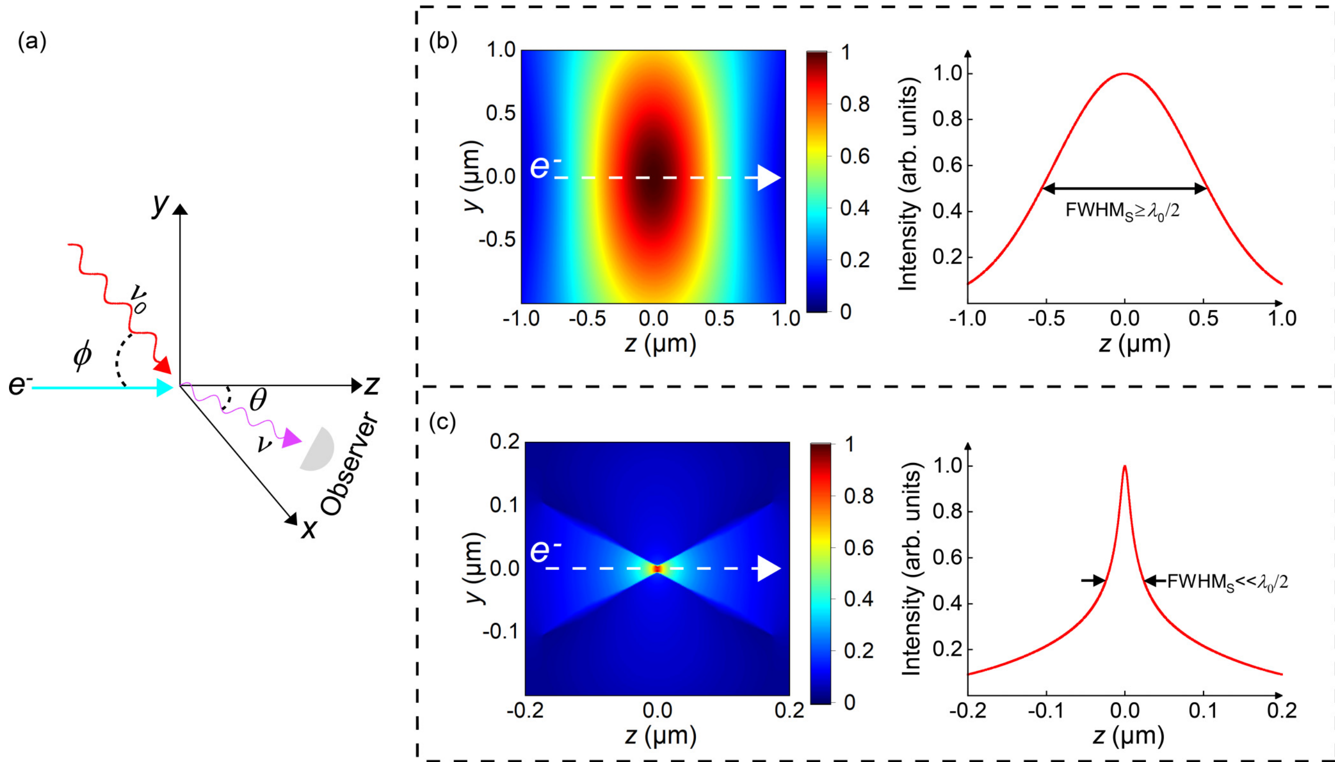

FIG. 1. Schematic illustration of breaking the half-cycle temporal limit of an optical pulse. (a) Inverse Compton scattering (ICS) between a photon and a relativistic electron. Typically, the electron energy is much higher than the initial photon energy, and the photon energy will be increased after ICS (i.e., $\nu > \nu_0$), while the energy loss of the electron is negligible. (b) Calculated field intensity distribution (left panel) and $\mathrm{FWHM_S}$ (right panel) of a conventional Gaussian-profile 1.8-μm-wavelength optical field that is focused on approaching the diffraction limit. Due to the diffraction limit, the $\mathrm{FWHM_S}$ cannot be smaller than $\lambda_0/2$, which sets the lower limit of the transition time (τ) to be larger than half an optical cycle when an electron passes through. (c) Calculated field intensity distribution (left panel) and $\mathrm{FWHM_S}$ (right panel) of a deep-sub-wavelength confined 1.8-μm-wavelength optical field obtained in a SCS composed of two 500-nm-diameter cadmium sulfide (CdS) nanowires, with a slit width of 20 nm. Here the $\mathrm{FWHM_S}$ can be much smaller than $\lambda_0/2$, making it possible to decrease τ well below half an optical cycle, and generate a deep-sub-cycle optical pulse that breaks the half-cycle temporal limit.

(energy of E_e, speed of $\boldsymbol{V}_\mathbf{e}$) with a much higher energy, it can be converted into a photon with a higher energy (i.e., a higher frequency ν and a shorter wavelength λ) by ICS, in which ν is dependent on the incident angle ϕ and the scattering angle θ [22,23]

$$\nu(\phi, \theta) = \nu_0 \frac{1 - \beta \cos\phi}{1 - \beta \cos\theta}, \tag{1}$$

where $\boldsymbol{\beta} = \boldsymbol{V}_\mathbf{e}/c$. In this work, $\phi = 90^\circ$, therefore

$$\nu(\theta) = \frac{\nu_0}{1 - \beta \cos\theta}, \tag{2}$$

or equivalently $\lambda = \lambda_0(1 - \beta\cos\theta)$.

Usually, the photon is provided by a focused laser beam in free space with a spot size larger than the diffraction limit [Fig. 1(b)], as has been reported in previous works [24–26]. When an electron with a velocity V_e passes through the focused optical field with a Gaussian-profile intensity distribution ($\mathrm{FWHM_S} \geqslant \lambda_0/2$), the scattered photons will compose a virtual pulse (compared to a typical pulse that usually contains a large number of photons, the single-electron-scattered pulse typically has an average photon number much less than 1, and is thus called a "virtual" pulse here) with a widthτ that can be estimated by

$$\tau \approx \left(\frac{\mathrm{FWHM_S}}{V_e}\right)(1 - \beta\cos\theta), \tag{3}$$

where θ is the emission angle [see Fig. 1(a)], $\beta = V_e/c$, and the factor $(1 - \beta\cos\theta)$ comes from the time compression effect for the observer facing the scattered photon [27]. Since $\lambda = \lambda_0 \cdot (1 - \beta\cos\theta)$, $V_e < c$, we have $\tau > \lambda/2c = T_c/2$, showing that the width of a pulse composed of conventional ICS photons can not be shorter than the half-cycle limit of $T_c/2$.

Recently, a deep-sub-wavelength-confined optical field has been realized in a slit waveguide mode supported by a coupled-nanowire-pair slit confinement structure (SCS) [28–30], which can offer a high-intensity peak field with $\mathrm{FWHM_S} \ll \lambda_0/2$ at the center of the mode in the optical near field (Fig. 1(c); see also Sec. 1.1 in the Supplemental Material [31]), and a low-intensity diffraction-limited background field with a peak-to-background ratio up to 30 dB. When an electron passes through such a field, a deep-sub-cycle virtual pulse can be obtained with a temporal width $\tau < T_c/2$, provided that $V_e > 2c \cdot \mathrm{FWHM_S}/\lambda_0$. For example, if $\mathrm{FWHM_S}/\lambda_0 = 1/4$, to have $\tau < T_c/2$, the electron should have $V_e > 0.5c$, which can

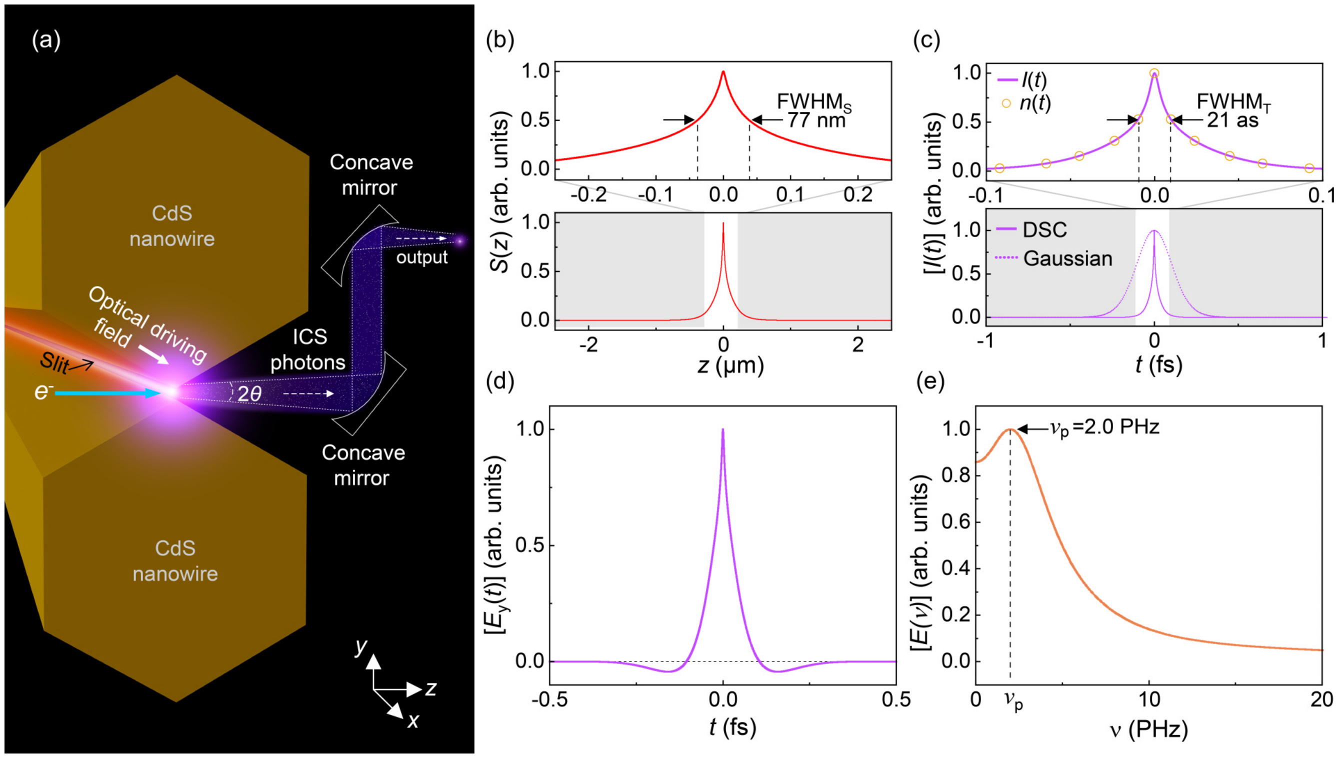

FIG. 2. Generation of deep-sub-cycle optical pulses with single-electron incidence. (a) Schematic diagram of the ICS system. An electron incident from the left side collides perpendicularly with photons in a confined optical driving field, which is generated along a slit between two coupled CdS nanowires. (b) Normalized z-axis Poynting vector of an optical driving field, generated with a slit size of 20 nm, a nanowire diameter of 500 nm, and a wavelength of 1.8 μm. The close-up image (upper panel) shows the extremely confined field with an $\mathrm{FWHM_S}$ of 77 nm. (c) $I(t)$ of the virtual ICS pulse (purple solid line), giving an $\mathrm{FWHM_T}$ of 21 as (upper panel). For reference, photon numbers $n(t)$ is also provided (orange circle), which basically overlaps with $I(t)$. For comparison, $I(t)$ of a virtual ICS pulse generated in a Gaussian-profile optical driving field is also provided (purple dotted line, lower panel). DSC: deep-sub-cycle pulse, Gaussian: Gaussian-profile pulse. (d) $E_y(t)$ of the DSC pulse in (c). (e) Fourier spectrum $E(\nu)$ of the pulse in (d), giving a peak frequency of $\sim$2.0 PHz.

be readily achieved by relatively small electron acceleration systems (e.g., an electron microscope [32]).

III. DEEP-SUB-CYCLE VIRTUAL PULSE OBTAINED WITH SINGLE-ELECTRON INCIDENCE

For principle exploration, we first consider the simplest case with single-electron incidence. We assume that the wavelength of the pulsed optical driving field (λ_0) is 1.8 μm (a common wavelength of the driving field in high harmonic generation (HHG) [33,34]), and the field is confined as a slit waveguide mode formed by coupling two identical cadmium sulfide (CdS) single-crystal nanowires [i.e., a coupled-nanowire-pair SCS, see Fig. 2(a)] [28,29]. By assuming a slit width of 20 nm and a nanowire diameter of 500 nm, we have an $\mathrm{FWHM_S} \sim 77$ nm (corresponding to an $\mathrm{FWHM_S}/\lambda_0 \sim 0.04$) of the central peak along the z direction [Fig. 2(b)]. Meanwhile, we assume that the electron has a kinetic energy of 1 MeV (corresponding to a velocity $V_e \sim 0.94c$), and passes through the confined optical field from left to right along the central symmetry line (i.e., the z direction) of the SCS cross section [Fig. 2(a)]. Due to the symmetry of the SCS structure and thus that of the optical driving field, the electron collides with photons almost perpendicularly on their motion trajectories [i.e., $\phi = \pi/2$ in Fig. 1(a)]. In such a case, the path-dependent (i.e., z dependent) scattered photon number $n(z)$ can be obtained as [24] (see also Sec. 1.2 in Supplemental Material for details [31])

$$n(z) = N_0 N_e \sigma_T \frac{S(z)}{P_{\mathrm{mode}}} \frac{\delta_t}{\tau_0}, \tag{4}$$

where the total photon number of the driving field $N_0 \approx E_{p0}/h\nu_0$, in which E_{p0} is the total energy of the driving pulse, h is the Planck constant, and ν_0 the central frequency of the driving pulse; N_e is the number of incident electrons (here $N_e = 1$), σ_T is the Thomson scattering cross section ($\sigma_T = 6.65 \times 10^{-29}\ \mathrm{m}^2$), $S(z)$ is the z-dependent Poynting vector, P_{mode} is the mode power (i.e., the mode area integral of the Poynting vector) of the optical driving field at the moment the electron passing through, δ_t is the transition time of the electron, and τ_0 is the width of the driving pulse (with an assumption of $\tau_0 > \delta_t$). Since δ_t (typically tens to hundreds of attoseconds) is much shorter than τ_0 (typically tens to hundreds of femtoseconds), P_{mode} is almost a constant during the time of the electron transit, and can be estimated as $P_{\mathrm{mode}} = E_{p0}/\tau_0$.

When the electron passes through the driving field along the z direction [Fig. 2(b)], due to the negligible energy loss of the electron compared with its initial energy (i.e., 1 MeV), V_e can be assumed as a constant (i.e., 0.94c) and the trajectory of the electron is almost a straight line (see also Sec. 1. 4 in the Supplemental Material for details [31]). Meanwhile, since the electron-photon-interaction area (i.e., the spot for generating the ICS pulse) is comparable to the wavelength of the ICS photons, the ICS pulse will have a spherical wave front when it propagates to the spatial far field. Thus, we can assume a spherical receiving surface centered on the center of the slit with a radius of r, a scattering angle of θ, and an azimuthal angle of ψ (see also Fig. S2), or equivalently, a concave mirror group to guide the scattered photons out and focused it back into a point in the far field with an almost equal optical path for photons generated with different θ [Fig. 2(a)]. Thus, the number of scattered photons received at time t and position $Q(r, \theta, \psi)$ can be approximately obtained as [24]

$$n(t, \theta, \psi) = n(t) \cdot f(\theta, \psi), \tag{5}$$

where the path dependence $n(z)$ (i.e., z dependence) is converted to the time dependence $n(t)$ (i.e., t dependence) by

$$t = \frac{z}{V_e}(1 - \beta\cos\theta) + \frac{r}{\mathrm{c}}, \tag{6}$$

and $f(\theta, \psi)$ describes the angular distribution of scattered photons at the receiving surface [24]

$$f(\theta, \psi) = K\chi^2\left[\tfrac{1}{2} + \chi(\chi - 1)(1 + \cos 2\psi)\right], \tag{7}$$

where K is a constant coefficient and $\chi = 1/[1 + \gamma^2\theta^2]$.

By integrating Eq. (5) over the azimuthal angle ψ and incorporating with Eq. (2), the intensity of the ICS pulse at time t and angle θ can be obtained as

$$I(t, \theta) = n(t, \theta)h\nu(\theta). \tag{8}$$

By integrating Eq. (8) over all possible θ, we obtain the temporal profile of the virtual pulse [i.e., I(t)] with a temporal full width at half maximum ($\mathrm{FWHM_T}$, i.e., τ) of 21 as [Fig. 2(c)], which is much narrower than that of an ICS pulse generated with the same electron but a 1.8-μm-wavelength diffraction-limited driving field with a conventional Gaussian-profile intensity distribution [purple dotted line in Fig. 2(c)].

Since the phase of the scattered field is inherited from the optical driving field [35], to maximize the peak intensity of the ICS pulse, we assume that the amplitude of the driving field reaches the positive maximum when the electron arrives at the central point of the slit (i.e., $z = 0$). Thus, in the spatial far field, the transverse scattered electric field $E_y(t)$ can be obtained as (see also section 1.3 in the Supplemental Material for details [31]):

$$E_y(t, \theta) = \sqrt{2\sqrt{\frac{\mu_0}{\varepsilon_0}} I(t, \theta)}\, e^{i2\pi\nu_0\left(\frac{t}{1-\beta\cos\theta}\right)}, \tag{9}$$

where ε_0 and μ_0 are the vacuum permittivity and the magnetic permeability, respectively. By integrating over all possible θ, we obtain $E_y(t)$ shown in Fig. 2(d), which agrees well with the classical radiation model of an arbitrarily accelerated electron (see also Sec. 1. 5 in Supplemental Material for details [31]). Unlike the conventional dipolar pulses, here the dominant peak is generated by the unidirectional acceleration of the electron driven by the optical driving field, and is thus a unipolar pulse with nonzero electric area [36–42]. Figure 2(e) gives the Fourier spectrum of such a virtual pulse, revealing a peak frequency ν_p of 2 PHz (corresponding to a $T_c \sim 0.5$ fs). After the peak point ν_p, the amplitude of $E(\nu)$ decreases quickly with increasing frequency. As $\mathrm{FWHM_T}$ (21 as) is much shorter than $T_c/2$ (i.e., 0.25 fs), such an ICS-generated pulse is a deep-sub-cycle pulse.

It is worth mentioning that, compared with the temporal profiles of previously reported ultrafast subcycle or multiple-cycle pulses [14,16,24], here the deep-sub-cycle pulse has a spikelike profile that is much sharper at the center, which enables a much tighter temporal confinement and a much larger temporal gradient for achieving the temporal super-resolution in the "temporal near field" (i.e., the time region which is much less than one cycle away from $t = 0$, see section "Temporal near field" in the following text). For ease of comparison, we define a temporal confinement factor $\zeta = T_c/\tau$, in which $\zeta > 2$ sets a deep-sub-cycle criterion that breaks the conventional half-cycle limit. For example, for the virtual pulse mentioned above, the calculated $\zeta \sim 24$.

IV. DEEP-SUB-CYCLE ULTRAFAST PULSE OBTAINED WITH ELECTRON-BUNCH INCIDENCE

Using single-electron incidence (i.e., $N_e = 1$), the photon number of a single ICS pulse (N_{SP}) is too small ($\ll 1$) to compose a real pulse with $N_{SP} \gg 1$. For reference, when using a power density of the optical driving field approaching the damage threshold of the CdS crystal ($\sim 1.7 \times 10^{16}\ \mathrm{W/m^2}$ [43,44]), the calculated N_{SP} is $\sim 5.4 \times 10^{-9}$. To increase N_{SP}, we propose to use a high-density ultrafast electron bunch [45–49] (i.e., to increase the electron number), optical material with a higher optical damage threshold to construct the SCS (i.e., to increase the photon number in the optical driving field), driving field with a longer wavelength and thus a larger slit size (i.e., increase the ICS volume and thus the electron and photon numbers involved). Meanwhile, to maintain a short transition time δ_t of the electron in the enlarged slit size, we use an electron bunch with higher electron energy and thus a larger time compression effect. Specifically, we use the following parameters: (1) an electron bunch with a duration of 1 fs (satisfying a Gaussian distribution), an electron energy of 3 MeV, an elliptical cross section of 25 μm and 125 μm for the short and the long axes, respectively, and an electron charge of 0.1 nC (corresponding to an electron density of $8.6 \times 10^{23}/\mathrm{m^3}$); (2) a pulsed driving field with a central frequency of 0.4 THz and a pulse width of 1 ps, which has been experimentally realized with a single-pulse energy up to 1.4 mJ [50]; (3) a SCS consisted of two identical sapphire hexagonal prisms [Fig. 3(a)], with a diagonal diameter of 125 μm and a slit size of 25 μm (corresponding to an $\mathrm{FWHM_S}$ of 62.8 μm). By using $\sim 5\%$ energy of the 1.4-mJ terahertz pulses [50] within a 0.05-THz bandwidth around the 0.4-THz central frequency, we have a power density of $3.8 \times 10^{15}\ \mathrm{W/m^2}$ at the center of the slit (see Fig. S8 in Sec. 2 of Supplemental Material), which is far below the optical damage threshold of $2 \times 10^{18}\ \mathrm{W/m^2}$ of the sapphire [51]. In addition, similar to the single-electron-incidence case, the energy loss of the electron bunch is negligible during the ICS process.

With the above assumptions, we obtained an ICS pulse shown in Fig. 3(b), which is composed of a series of single-electron-incidence subpulses (see Fig. S4 in Sec. 1. 2 of the Supplemental Material [31]). Owing to the significantly increased electron number (from $N_e = 1$ to $\sim 6.2 \times 10^8$) and

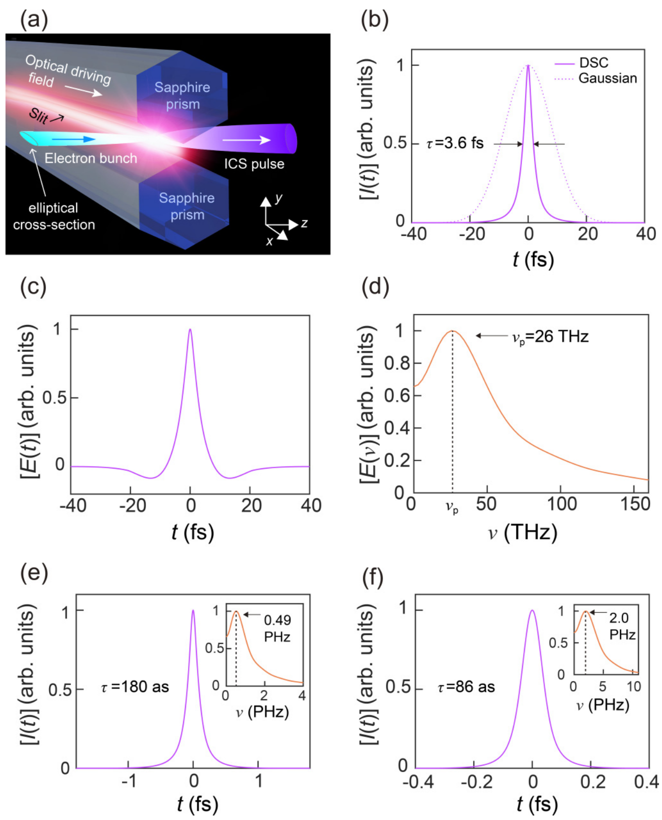

FIG. 3. Generation of deep-sub-cycle optical pulses with electron-bunch incidence in a THz optical driving field. (a) Schematic diagram of the ICS system. A focused electron bunch with an elliptical cross-section is incident from the left side of a slit, collides with photons in the confined optical driving field that is generated along the slit between two coupled sapphire prisms. (b) $I(t)$ of the ICS pulse, generated via ICS of a 1-fs 3-MeV electron bunch in a driving field with a $\mathrm{FWHM_S}$ of 62.8 μm, a frequency of 0.4 THz, and a pulse width of 1 ps. The measured pulse width (τ) is 3.6 fs. For comparison, $I(t)$ of an ICS pulse generated in a 0.4-THz-frequency Gaussian-profile optical driving field is also provided (purple dotted line). DSC: deep-sub-cycle pulse, Gaussian: pulse obtained with a Gaussian-profile driving field. (c) $E_y(t)$ of the DSC pulse in (b). (d) Fourier spectrum $E(\nu)$ of the pulse in (c), giving a peak frequency of ∼26 THz. (e) $I(t)$ of an ICS pulse generated with a 50-as 15-MeV electron bunch, offering a τ of ∼180 as. Inset, $E(\nu)$ with a ν_p of ∼0.49 PHz. (f) $I(t)$ of an ICS pulse generated with a 50-as 30-MeV electron bunch, offering a τ of ∼86 as. Inset, $E(\nu)$ with a ν_p of ∼2.0 PHz. Driving fields used in (e) and (f) are the same as in (b).

the interacting driving-field photon number, the N_{SP} of the ICS pulse is increased to $\sim 2.3\times10^5$. Meanwhile, despite the temporal broadening by the pulse width of the electron bunch (i.e., $\tau_e = 1$ fs), the overall ICS-pulse width τ (i.e., $\mathrm{FWHM_T}$) of 3.6 fs, remains much narrower than that of an ICS pulse generated with the same electron bunch but a 0.4-THz-frequency diffraction-limited driving field with a Gaussian-profile intensity distribution [purple dotted line in Fig. 3(b)]. Figure 3(c) gives calculated $E_y(t)$ of the ICS pulse shown in Fig. 3(b), which retains an evident unipolar feature, similar to that of a single-electron-incidence ICS pulse [Fig. 2(d)]. Figure 3(d) gives $E(\nu)$, the Fourier transform of the $E_y(t)$ in Fig. 3(c), revealing a peak frequency ν_p of ∼26 THz (corresponding to a $T_c\sim 38$ fs and $\lambda\sim11.5$ μm) and a temporal confinement factor ζ (i.e., T_c/τ) of 10.6 (corresponding to a 0.095-cycle 3.6-fs-width midinfrared femtosecond pulse).

By using an electron bunch with higher energy and shorter pulse width, we obtain deep-sub-cycle attosecond pulses with ν_p in the visible and UV spectral ranges. Figure 3(e) gives $I(t)$ of an attosecond pulse generated using a 15-MeV 50-as 50-pC electron bunch (maintaining an almost the same electron density of $8.5\times10^{23}/\mathrm{m}^3$) and the same 0.4-THz driving field, offering a ν_p of ∼0.49 PHz (corresponding to a $T_c\sim 2.0$ fs and λ∼620 nm), a τ of ∼180 as and a ζ of ∼11, corresponding to a "visible" 0.091-cycle 180-as-width attosecond pulse. Figure 3(f) gives $I(t)$ of an attosecond pulse obtained using a 30-MeV 50-as 50-pC electron bunch and the same 0.4-THz driving field, with $\nu_p\sim 2.0$ PHz (corresponding to a $T_c\sim 499$ as and $\lambda\sim150$ nm), $\tau\sim86$ as and $\zeta\sim5.8$, corresponding to an "ultraviolet" 0.17-cycle 86-as-width attosecond pulse. Due to the reduced pulse width of the electron bunch and thus the total electron number involved, the single-pulse photon number N_{SP} is reduced to $\sim 1.1\times10^4$, however, it can be increased by using a higher-intensity driving field, higher-density electron bunch, or reducing ζ to a certain value larger than 2.

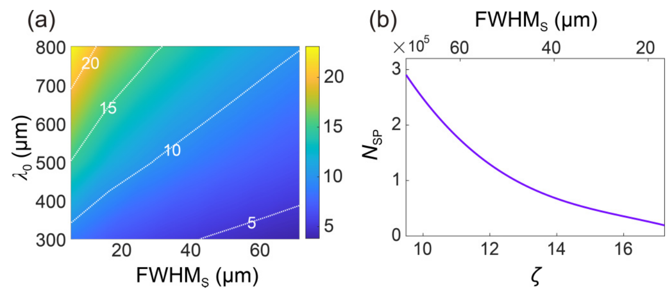

FIG. 4. Trade off between ζ and N_{SP}. (a) Dependence of ζ on $\mathrm{FWHM_S}$ and λ_0 of the optical driving field, with a 3-MeV 1-fs electron bunch. (b) Dependence of N_{SP} on ζ and $\mathrm{FWHM_S}$, with a λ_0 of 750 μm.

V. TRADE-OFF BETWEEN ζ AND N_{SP}

By changing the wavelength (i.e., λ_0) and/or the confinement (i.e., $\mathrm{FWHM_S}$) of the optical driving field, it is possible to change ζ within a wide range. For reference, Fig. 4(a) gives the dependence of ζ on λ_0 and $\mathrm{FWHM_S}$ of the driving field, with the same 3-MeV electron bunch used in Fig. 3(b). It shows that, when λ_0 is increased from 300 to 800 μm, ζ can be increased from $\sim$ 4 to larger than 20. With the same λ_0, ζ can be increased by reducing $\mathrm{FWHM_S}$. However, with increasing ζ, N_{SP} decreases [Fig. 4(b)]. For reference, when ζ is increased from 10 to 16, N_{SP} decreases from 2.5×10^5 to 3.5×10^4. Therefore, there's a trade-off between ζ and N_{SP}.

VI. TEMPORAL NEAR FIELD

Similar to the optical near field in the spatial scale that is localized around the origin of the spatial coordinate (i.e., $z = 0$) and has a higher spatial gradient (i.e., $\mathrm{d}E/\mathrm{d}z$ along the z direction), here we define an equivalent concept called "temporal near field", an optical near field in the temporal scale, which is localized around the origin of the temporal

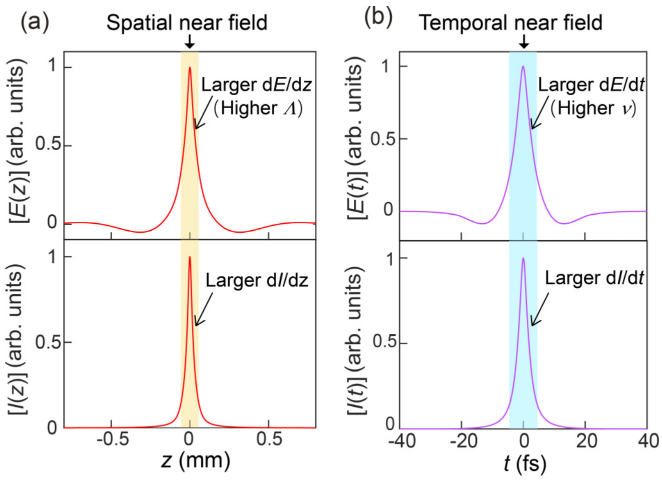

FIG. 5. Spatial and temporal near field. (a) Calculated $E(z)$ and $I(z)$ of optical driving field used in Fig. 3, which has a intensity $\mathrm{FWHM_S}$ of 62.8 μm, a frequency of 0.4 THz and a pulse width of 1 ps. The light orange area indicates the spatial near field localized around $z = 0$. (b) Calculated $E(t)$ and $I(t)$ of the deep-sub-cycle pulse shown in Fig. 3(c) (upper) and Fig. 3(b) (bottom), which is generated by the optical driving field in (a) and a 3-MeV 1-fs electron bunch. The light blue area indicates the temporal near field localized around $t = 0$.

coordinate (i.e., $t = 0$) and has a higher temporal gradient (i.e., dE/dt along the t direction), as shown in Fig. 5.

Also, similar to the spatial near field, in which the higher spatial gradient can offer higher spatial frequency (Λ) for achieving higher spatial resolution in applications such as near-field optical microscopy [Fig. 5(a)] [1], the higher temporal gradient can offer higher temporal frequency (i.e., the light frequency ν) for achieving higher temporal resolution in resolving ultrafast dynamics in light-matter interaction [Fig. 5(b)].

Moreover, similar to the spatial near field, which can only be manifested in a spatial confined field with sub-half-wavelength feature, the temporal near field can only be manifested in a temporal confined-field with sub-half-cycle feature, as the deep-sub-cycle ultrafast optical pulse we introduced here.

VII. TEMPORAL EVOLUTION OF A DEEP-SUB-CYCLE PULSE IN FREE SPACE

After its generation, the ICS pulse will propagate from the slit region to spatial far-field free space with a spherical wave front around the z axis within a certain solid angle [Fig. 6(a)], followed by a slower and much less divergent electron bunch (see also Fig. S10 in Supplemental Material for details [31]). The electric field $E_y(t)$, including the far-field ICS radiation field and the near-field electrostatic field in the spatial scale, can be obtained by Ref. [27]

$$E_y(t) = \frac{e}{4\pi\varepsilon_0}\left\{\frac{[\hat{\boldsymbol{r}}\times[(\hat{\boldsymbol{r}}-\boldsymbol{\beta})\times\dot{\boldsymbol{\beta}}]]}{cr(1-\hat{\boldsymbol{r}}\cdot\boldsymbol{\beta})^3}+\frac{(1-\beta^2)(\hat{\boldsymbol{r}}-\boldsymbol{\beta})}{r^2(1-\hat{\boldsymbol{r}}\cdot\boldsymbol{\beta})^3}\right\}, \tag{10}$$

where $\hat{\boldsymbol{r}}$ is the unit vector pointing from the electron to the receiving surface, $\boldsymbol{\beta} = \boldsymbol{V}_\mathbf{e}/c$, and $\dot{\boldsymbol{\beta}}$ is the acceleration of electrons. With Eq. (10), we calculate $E_y(t)$ and carrier envelope (CE) of an ICS pulse generated with a 0.4-THz driving field and a 3-MeV 1-fs electron bunch [the same parameters in Fig. 3(b)], received on a spherical surface after different propagation distances, as shown in Figs. 6(b)–6(d). When the propagation distance (i.e., the value of z) is small, e.g., $z = 0.5$ cm, $E_y(t)$ and CE have an obvious asymmetric profile [Fig. 6(b)], due to the non-negligible contribution from the asymmetric electrostatic field of the electron bunch. When we increase z to1 cm, compared to that of the ICS field (decays at a rate of $1/z$), the amplitude of the electrostatic field (decays at a rate of $1/z^2$) reduces to a negligible level, and $E_y(t)$ and CE becomes more symmetric [Fig. 6(c)]. When we further increase z to 10 cm, the electrostatic field is completely negligible, $E_y(t)$ and CE have stable symmetric profiles [Fig. 6(d)], showing a stable CE of the deep-sub-cycle ICS pulse propagating in spatial far-field free space. For reference, Fig. 6(e) gives the calculated z-dependent amplitude of the near-field part of $|E_y|$ [i.e., the second term in Eq. (10)], clearly showing its quadratic decay behavior.

VIII. EXPERIMENTAL FEASIBILITY

Although the experimental implementation of the above-mentioned proposal seems challenging at present, based on the state-of-the-art experimental results in this field, it will be experimentally feasible in the near future. Here we discuss experimental feasibility from several key aspects.

Firstly, to avoid optical damage of the sapphire prism around the slit side (where the power density goes to the maximum; see Fig. S8 in Sec. 2 of Supplemental Material), in the calculation we assume a peak density of $3.8 \times 10^{15}\ \mathrm{W/m^2}$ around the slit, which is about 500 times lower than the damage threshold experimentally measured in fine polished sapphire crystals ($2 \times 10^{18}\ \mathrm{W/m^2}$ [51]), and is low enough to avoid the generation of "hot spots" in the sapphire prism.

Secondly, the 3-MeV 1-fs 0.1-nC electron bunch we used in Fig. 3(b) for generating the deep-sub-cycle femtosecond pulse, has been recently experimentally realized [52]. Meanwhile, the 50-as-width electron bunch has been experimentally achieved [47], but the electron energy (currently hundreds of eV) needs to be further accelerated to the MeV level to generate the deep-sub-cycle attosecond pulse shown in Figs. 3(e) and 3(f).

Also, in the experiments, electrons in the bunch will have a certain degree of energy dispersion. As a reasonable estimation, we assume a 5% energy disperion of the 3-MeV 1-fs 0.1-nC electron bunch employed in Fig. 3(b), and calculate the influence of the energy dispersion on the pulse width of the generated deep-sub-cycle femtosecond pulse (see Fig. S10 in Supplemental Material for details [31]). It shows that, compared with that obtained with a zero-dispersion electron bunch, the ICS pulse width increases from 3.6 fs to 3.9 fs, corresponding to a reduction of ζ from 10.6 to 8.8 (see also Fig. S11 in Supplemental Material for details [31]). As ζ remains much larger than 2, the sub-half-cycle feature can be well maintained when using electron bunches with a certain energy dispersion.

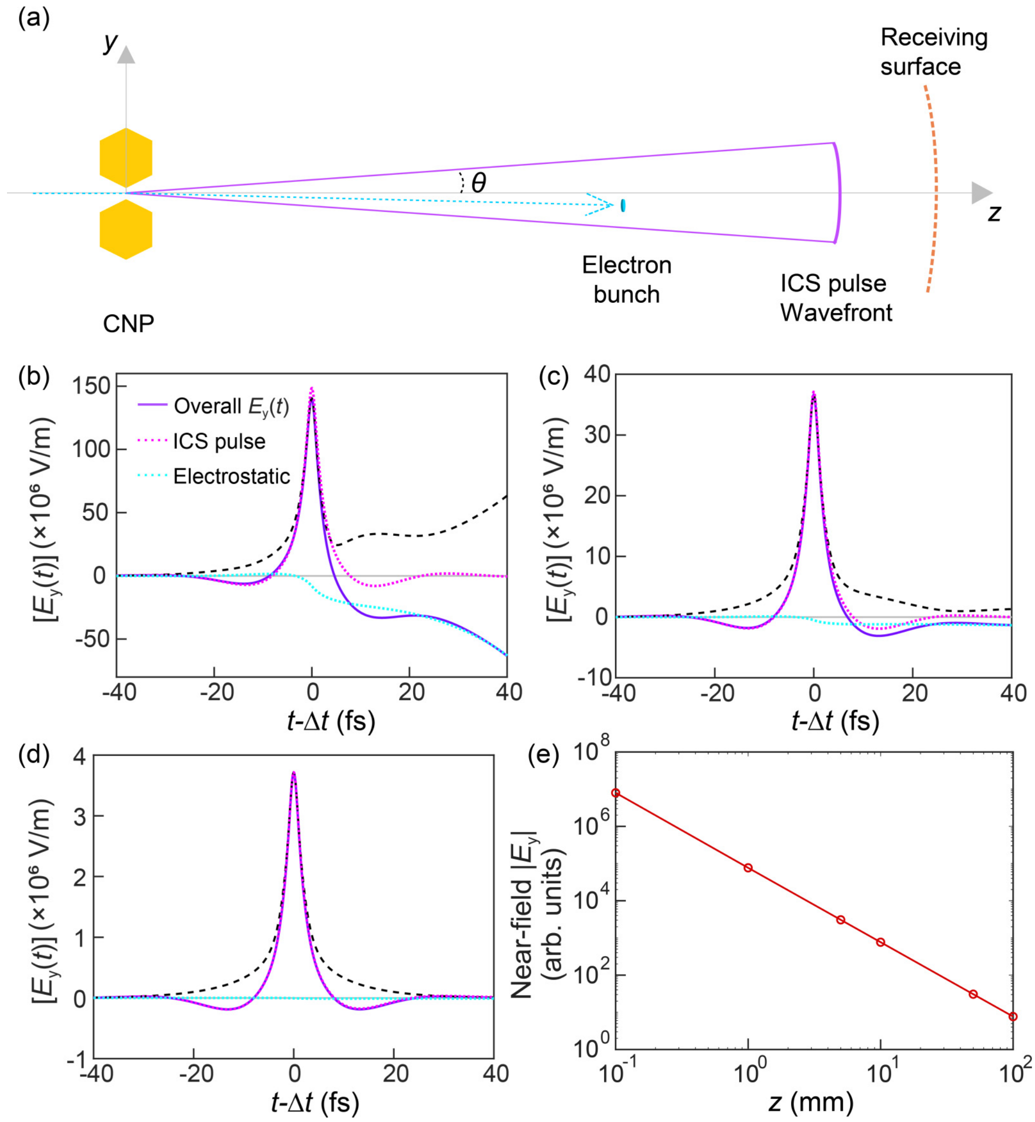

FIG. 6. Temporal electric field evolution of an ICS pulse. (a) Schematic illustration of an ICS pulse propagating from the 0.4-THz driving field of the SCS to far-field free space (not to scale). At the far field, the ICS pulse has a spherical wavefront (purple arc) propagating at light speed, followed by a slower and less divergent electron bunch (cyan sheet) with a small deflected angle. The spherical receiving surface (orange dashed arc) can be shifted along the z axis to change the propagation distance of the ICS pulse. (b)–(d) Temporal evolution and CE of the ICS pulse during propagation. Here the overall transverse electric field $E_y(t)$ (purple solid line) is consisted of the ICS pulse (magenta dotted line) and the electrostatic field of the electron (cyan dotted line), with a propagation distance z assumed to be (b) $z = 0.5$ cm, (c) $z = 1$ cm, and (d) $z = 10$ cm. Calculated CE of $E_y(t)$ is also given (black dashed line). $\Delta t = \Delta z/c$ is the pulse propagation time from the slit center (i.e., $z = 0$) to the receiving surface. (e) Calculated z-dependent amplitude of the near-field part of $|E_y|$.

Finally, when the electron bunch passes through the slit of the sapphire-prism pair, besides the ICS process, there're other possible radiation processes (e.g., Cherenkov or Smith-Purcell radiation) [53] originated from the electromagnetic polarization of the sapphire prisms by the transient electron field. As an estimation, we calculate the overall radiation from the polarization of the sapphire prisms, and compare it with the ICS pulse (see Fig. S12 in Supplemental Material for details [31]). It shows that, due to the symmetrical polarization of the two sapphire prisms, in the spatial far field, the fields of dielectric radiation from the two prisms cancel each other out, leaving a radiation field more than three orders of magnitude lower in electric field than that of the ICS electric field. Also, due to the longer optical path involved, the prism radiation lags about several picoseconds behind the ICS pulse. Therefore, compared with the ICS pulse, the abovementioned other radiation can be neglected.

IX. CONCLUSION

So far, we have demonstrated an approach to generating deep-sub-cycle ultrafast optical pulses. Compared with existing methods for subcycle ultrafast pulse generation (e.g., OPCPA, OAWG, and RES) that have not yet reached the half-cycle limit, our approach shown here can break the half-cycle limit and achieve deep-sub-cycle pulses with $\zeta > 2$

(see Sec. 4 and Fig. S11 in Supplemental Material [31]), while maintaining a stable CE when propagating into the far-field free space. Also, the single-pulse photon number of the deep-sub-cycle can be significantly increased when higher-intensity THz pulse and/or higher-density relativistic electron bunch are available. As dipolar/multipolar oscillation in optical polarization or resonance is, in general, the basic mechanism of light-matter interaction [54], and the half-cycle pulse width is the temporal resolution limit of the current ultrafast optical technology [39,55], the deep-sub-cycle optical pulse demonstrated here can not only break the half-cycle limit and reveal deep-sub-cycle dynamics in light-matter interaction, but also manipulate (e.g., excite or mediate) optical response on the deep-sub-cycle level and generate previously difficult-to-reach states or processes (e.g., nondipole photoionization [56], anharmonic or temporal dark states), and pave a way to temporal super-resolution optical technology ranging from deep-sub-cycle ultrafast microscopy, spectroscopy to atom/molecule manipulation.

ACKNOWLEDGMENTS

We thank Y. R. Shen for helpful discussions. This work was supported by the National Key Research and Development Program of China (2024YFA1209200), the New Cornerstone Science Foundation (NCI202216) and the National Natural Science Foundation of China (62175213)

DATA AVAILABILITY

Some of the data that support the findings of this article are openly available [69], embargo periods may apply. Other data are available from the authors upon reasonable request.

[1] E. Betzig, J. K. Trautman, T. D. Harris, J. S. Weiner, and R. L. Kostelak, Breaking the diffraction barrier: Optical microscopy on a nanometric scale, Science **251**, 1468 (1991).

[2] S. W. Hell *et al.*, The 2015 super-resolution microscopy roadmap, J. Phys. D **48**, 443001 (2015).

[3] A. H. Zewail, Four-dimensional electron microscopy, Science **328**, 187 (2010).

[4] R. F. Egerton and M. Watanabe, Spatial resolution in transmission electron microscopy, Micron **160**, 103304 (2022).

[5] A. H. Zewail, Femtochemistry: Atomic-scale dynamics of the chemical bond, J. Phys. Chem. A **104**, 5660 (2000).

[6] A. Stolow, A. E. Bragg, and D. M. Neumark, Femtosecond time-resolved photoelectron spectroscopy, Chem. Rev. **104**, 1719 (2004).

[7] P. B. Corkum and F. Krausz, Attosecond science, Nat. Phys. **3**, 381 (2007).

[8] F. Krausz and M. Ivanov, Attosecond physics, Rev. Mod. Phys. **81**, 163 (2009).

[9] J. Itatani, F. Quéré, G. L. Yudin, M. Y. Ivanov, F. Krausz, and P. B. Corkum, Attosecond streak camera, Phys. Rev. Lett. **88**, 173903 (2002).

[10] D. M. Jonas, Two-dimensional femtosecond spectroscopy, Annu. Rev. Phys. Chem. **54**, 425 (2003).

[11] A. Sakdinawat and D. Attwood, Nanoscale x-ray imaging, Nat. Photon. **4**, 840 (2010).

[12] R. C. Dunn, Near-field scanning optical microscopy, Chem. Rev. **99**, 2891 (1999).

[13] X. Chen, D. Hu, R. Mescall, G. You, D. N. Basov, Q. Dai, and M. Liu, Modern scattering-type scanning near-field optical microscopy for advanced material research, Adv. Mater. **31**, 1804774 (2019).

[14] S.-W. Huang *et al.*, High-energy pulse synthesis with sub-cycle waveform control for strong-field physics, Nat. Photon. **5**, 475 (2011).

[15] G. M. Rossi *et al.*, Sub-cycle millijoule-level parametric waveform synthesizer for attosecond science, Nat. Photon. **14**, 629 (2020).

[16] S. T. Cundiff and A. M. Weiner, Optical arbitrary waveform generation, Nat. Photon. **4**, 760 (2010).

[17] N. K. Fontaine, R. P. Scott, L. Zhou, F. M. Soares, J. Heritage, and S. Yoo, Real-time full-field arbitrary optical waveform measurement, Nat. Photon. **4**, 248 (2010).

[18] W. J. Ma *et al.*, Bright subcycle extreme ultraviolet bursts from a single dense relativistic electron sheet, Phys. Rev. Lett. **113**, 235002 (2014).

[19] E. Siminos, I. Thiele, and C. Olofsson, Laser Wakefield driven generation of isolated carrier-envelope-phase tunable intense subcycle pulses, Phys. Rev. Lett. **126**, 044801 (2021).

[20] C. Manzoni, O. D. Mücke, G. Cirmi, S. Fang, J. Moses, S. W. Huang, K. H. Hong, G. Cerullo, and F. X. Kärtner, Coherent pulse synthesis: Towards sub-cycle optical waveforms, Laser Photon. Rev. **9**, 129 (2015).

[21] C. Manzoni and G. Cerullo, Design criteria for ultrafast optical parametric amplifiers, J. Opt. **18**, 103501 (2016).

[22] G. R. Blumenthal and R. J. Gould, Bremsstrahlung, synchrotron radiation, and compton scattering of high-energy electrons traversing dilute gases, Rev. Mod. Phys. **42**, 237 (1970).

[23] G. Ghisellini, in *Radiative Processes in High Energy Astrophysics*, edited by G. Ghisellini, (Springer International Publishing, Heidelberg, 2013), pp. 65–88.

[24] R. W. Schoenlein *et al.*, Femtosecond x-ray pulses at 0.4 Å generated by 90° Thomson scattering: A tool for probing the structural dynamics of materials, Science **274**, 236 (1996).

[25] M. Placidi, S. Di Mitri, C. Pellegrini, and G. Penn, Compact FEL-driven inverse compton scattering gamma-ray source, Nucl. Instrum. Methods Phys. Res. Sect. A **855**, 55 (2017).

[26] J. Vyskočil, E. Gelfer, and O. Klimo, Inverse compton scattering from solid targets irradiated by ultra-short laser pulses in the 10^{22}–10^{23} W/cm^2 regime, Plasma Phys. Control. Fusion **62**, 064002 (2020).

[27] A. Hofmann, *The Physics of Synchrotron Radiation*, Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology (Cambridge University Press, Cambridge, 2004).

[28] H. Wu, L. Yang, P. Z. Xu, J. Gong, X. Guo, P. Wang, and L. M. Tong, Photonic nanolaser with extreme optical field confinement, Phys. Rev. Lett. **129**, 013902 (2022).

[29] L. Yang, Z. K. Zhou, H. Wu, H. L. Dang, Y. X. Yang, J. X. Gao, X. Guo, P. Wang, and L. M. Tong, Generating a sub-nanometer-

confined optical field in a nanoslit waveguiding mode, Adv. Photon. **5**, 046003 (2023).
[30] Y. R. Shen, Optical waveguiding along nanometer slits, Adv. Photon. **5**, 040503 (2023).
[31] See Supplemental Material at http://link.aps.org/supplemental/10.1103/s269-kl9z for (i) deep-sub-cycle pulses generation scheme and calculation methods; (ii) electron's trajectory and dispersion; (iii) comparison between classical and ICS methods; (iv) peak power density calculation; (v) other possible radiations, which includes Refs. [15,27–29,35,57–68].
[32] R. K. Li and P. Musumeci, Single-shot MeV transmission electron microscopy with picosecond temporal resolution, Phys. Rev. Appl. **2**, 024003 (2014).
[33] B. E. Schmidt, A. D. Shiner, C. Trallero-Herrero, H. J. Wörner, S. Patchkovskii, J. C. Kieffer, P. B. Corkum, D. M. Villeneuve, and F. Légaré, in *2011 Conference on Lasers and Electro-Optics Europe and 12th European Quantum Electronics Conference (CLEO/EUROPE-EQEC)* (IEEE, Munich, 2011), p. 1.
[34] G. Fazio, T. Balčiūnas, Z. Yin, D. Zimin, C. Brahms, J. Travers, and H. J. Wörner, in *The International Conference on Ultrafast Phenomena (UP) 2022*, edited by F. Légaré *et al.* (Optica Publishing Group, Montreal, Quebec, 2022), p. Th4A.38.
[35] A. L'Huillier, P. Balcou, K. J. Schafer, and K. C. Kulander, in *Coherence Phenomena in Atoms and Molecules in Laser Fields*, edited by A. D. Bandrauk and S. C. Wallace (Springer, Boston, MA, 1992), pp. 191–202.
[36] D. You, R. R. Jones, P. H. Bucksbaum, and D. R. Dykaar, Generation of high-power sub-single-cycle 500-fs electromagnetic pulses, Opt. Lett. **18**, 290 (1993).
[37] H. C. Wu and J. Meyer-ter-Vehn, Giant half-cycle attosecond pulses, Nat. Photon. **6**, 304 (2012).
[38] M. V. Arkhipov, R. M. Arkhipov, A. V. Pakhomov, I. V. Babushkin, A. Demircan, U. Morgner, and N. N. Rosanov, Generation of unipolar half-cycle pulses via unusual reflection of a single-cycle pulse from an optically thin metallic or dielectric layer, Opt. Lett. **42**, 2189 (2017).
[39] R. Arkhipov, M. Arkhipov, A. Pakhomov, I. Babushkin, and N. Rosanov, Half-cycle and unipolar pulses (Topical Review), Laser Phys. Lett. **19**, 043001 (2022).
[40] A. Pakhomov, M. Arkhipov, N. Rosanov, and R. Arkhipov, Ultrafast control of vibrational states of polar molecules with subcycle unipolar pulses, Phys. Rev. A **105**, 043103 (2022).
[41] N. Rosanov, Unipolar pulse of an electromagnetic field with uniform motion of a charge in a vacuum, Phys. Usp. **193**, 1127 (2023).
[42] N. N. Rosanov, M. V. Arkhipov, R. M. Arkhipov, and A. V. Pakhomov, Half-cycle electromagnetic pulses and pulse electric area, Contemp. Phys. **64**, 224 (2023).
[43] J. R. Bettis, A. H. Guenther, and R. A. House, Refractive-index dependence of pulsed-laser-induced damage, Opt. Lett. **4**, 256 (1979).
[44] C. You, S. Dai, P. Zhang, Y. Xu, Y. Wang, D. Xu, and R. Wang, Mid-infrared femtosecond laser-induced damages in As_2S_3 and As_2Se_3 chalcogenide glasses, Sci. Rep. **7**, 6497 (2017).
[45] M. Thévenet, A. Leblanc, S. Kahaly, H. Vincenti, A. Vernier, F. Quéré, and J. Faure, Vacuum laser acceleration of relativistic electrons using plasma mirror injectors, Nat. Phys. **12**, 355 (2016).
[46] Y. Morimoto, Attosecond electron-beam technology: A review of recent progress, Microscopy **72**, 2 (2023).
[47] H. Y. Kim, M. Garg, S. Mandal, L. Seiffert, T. Fennel, and E. Goulielmakis, Attosecond field emission, Nature (London) **613**, 662 (2023).
[48] F. J. García de Abajo and C. Ropers, Spatiotemporal electron beam focusing through parallel interactions with shaped optical fields, Phys. Rev. Lett. **130**, 246901 (2023).
[49] R. J. L. Timmis *et al.*, Attosecond and nano-Coulomb electron bunches via the zero vector potential mechanism, Sci. Rep. **14**, 10805 (2024).
[50] B. Zhang *et al.*, 1.4-mJ high energy terahertz radiation from lithium niobates, Laser Photon. Rev. **15**, 2000295 (2021).
[51] O. Uteza, B. Bussière, F. Canova, J. P. Chambaret, P. Delaporte, T. Itina, and M. Sentis, Laser-induced damage threshold of sapphire in nanosecond, picosecond and femtosecond regimes, Appl. Surf. Sci. **254**, 799 (2007).
[52] Y. Wan, S. Tata, O. Seemann, E. Y. Levine, S. Smartsev, E. Kroupp, and V. Malka, Femtosecond electron microscopy of relativistic electron bunches, Light Sci. Appl. **12**, 116 (2023).
[53] N. Rivera and I. Kaminer, Light–matter interactions with photonic quasiparticles, Nat. Rev. Phys. **2**, 538 (2020).
[54] Y. R. Shen, *Principles of Nonlinear Optics* (Wiley, New York, 1984).
[55] P. M. Paul, E. S. Toma, P. Breger, G. Mullot, F. Augé, P. Balcou, H. G. Muller, and P. Agostini, Observation of a train of attosecond pulses from high harmonic generation, Science **292**, 1689 (2001).
[56] J. Liang *et al.*, Attosecond-resolved nondipole photoionization dynamics, Nat. Photon. **18**, 311 (2024).
[57] Y. Yang, J. Gao, H. Wu, Z. Zhou, L. Yang, X. Guo, P. Wang, and L. Tong, Generating a nanoscale blade-like optical field in a coupled nanofiber pair, Photon. Res. **12**, 154 (2024).
[58] K. Lee, Y. H. Cha, M. S. Shin, B. H. Kim, and D. Kim, Relativistic nonlinear Thomson scattering as attosecond x-ray source, Phys. Rev. E **67**, 026502 (2003).
[59] M. Hentschel, R. Kienberger, C. Spielmann, G. A. Reider, N. Milosevic, T. Brabec, P. Corkum, U. Heinzmann, M. Drescher, and F. Krausz, Attosecond metrology, Nature (London) **414**, 509 (2001).
[60] G. Sansone *et al.*, Isolated single-cycle attosecond pulses, Science **314**, 443 (2006).
[61] E. Goulielmakis *et al.*, Single-cycle nonlinear optics, Science **320**, 1614 (2008).
[62] A. Wirth *et al.*, Synthesized light transients, Science **334**, 195 (2011).
[63] C. Manzoni, S.-W. Huang, G. Cirmi, P. Farinello, J. Moses, F. X. Kärtner, and G. Cerullo, Coherent synthesis of ultra-broadband optical parametric amplifiers, Opt. Lett. **37**, 1880 (2012).
[64] K. Zhao, Q. Zhang, M. Chini, Y. Wu, X. Wang, and Z. Chang, Tailoring a 67 attosecond pulse through advantageous phase-mismatch, Opt. Lett. **37**, 3891 (2012).
[65] J. Li *et al.*, 53-attosecond x-ray pulses reach the carbon K-edge, Nat. Commun. **8**, 186 (2017).
[66] T. Gaumnitz, A. Jain, Y. Pertot, M. Huppert, I. Jordan, F. Ardana-Lamas, and H. J. Wörner, Streaking of 43-attosecond soft-x-ray pulses generated by a passively CEP-stable mid-infrared driver, Opt. Express **25**, 27506 (2017).

[67] H. Liang *et al.*, High-energy mid-infrared sub-cycle pulse synthesis from a parametric amplifier, Nat. Commun. **8**, 141 (2017).

[68] M. Kretschmar, E. Svirplys, M. Volkov, T. Witting, T. Nagy, M. J. J. Vrakking, and B. Schütte, Compact realization of all-attosecond pump-probe spectroscopy, Sci. Adv. **10**, eadk9605 (2024).

[69] H. Dang, Dataset for "Deep-sub-cycle ultrafast optical pulses", Version v2 [Data set], Zenodo (2025), https://doi.org/10.5281/zenodo.15695623

Correction: A grant number in the Acknowledgments contained an error and has been fixed.